MTH2010

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

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Groups, Rings and Fields

Module Leader: Mohamed Saïdi

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No word count

SECTION A

All rings are commutative with an identity element 1.

1. (a) Let G be a group. Prove that the map $\phi : G \to G$ defined by

$$\phi(g) = g^2$$

is a group homomorphism if and only if G is abelian. (8)

(b) Prove that the centraliser of a 3-cycle in A_4 has cardinality 3.

What is the cardinality of a conjugacy class of A_4 which contains a 3-cycle?

What is the number of distinct conjugacy classes in A_4 containing a 3-cycle? (16)

(c) Let $\phi : R \to S$ be a ring homomorphism.

(i) Give the definition of the kernel $\mathrm{Ker}(\phi)$ of ϕ and prove it is an ideal of R. (8)

(ii) Let $u \in R^\times$ be a unit of R. Show that $\phi(u)$ is a unit of S and $\phi(u)^{-1} = \phi(u^{-1})$. (8)

(d) Let R be the ring

$$R = \mathbb{Z}[\sqrt{5}] = \{a + b\sqrt{5} : a, b \in \mathbb{Z}\}.$$

By exhibiting infinitely many units in R show that the group of units of R is infinite. (10)

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SECTION B

All rings are commutative with an identity element 1.

2. (a) Is there a surjective homomorphism from the dihedral group D_{12} with 12 elements to an abelian group of cardinality 4?

Is there a surjective homomorphism from D_{12} to a cyclic group of cardinality 4?

Justify your answers and construct such homomorphisms if they exists. (13)

(b) Let

$$\theta : \mathbb{Q}[X] \to \mathbb{Q}(i)$$

be the map defined by

$$\theta(a_0 + a_1X + \cdots + a_nX^n) = a_0 + a_1i + \cdots + a_ni^n.$$

Show that θ is a surjective ring homomorphism. Prove that $\mathrm{Ker}(\theta)$ is the ideal $(X^2+1)_{\mathbb{Q}[X]}$ of $\mathbb{Q}[X]$ generated by X^2+1. Deduce that the factor ring

$$\mathbb{Q}[X]/(X^2+1)_{\mathbb{Q}[X]}$$

is isomorphic to $\mathbb{Q}(i)$. State clearly any theorem you use. (12)

[**25**]

3. (a) Prove that a group of cardinality 56 must have a normal Sylow p-subgroup for some prime number p dividing 56. (10)

(b) Let $\phi : R \to S$ be a ring homomorphism.

(i) Prove that if P is a prime ideal of S then

$$\phi^{-1}(P) = \{r \in R : \phi(r) \in P\}$$

is a prime ideal of R. (7)

(ii) Prove that if M is a maximal ideal of S and ϕ is surjective then $\phi^{-1}(M)$ is a maximal ideal of R. Give an example to show that this need not be the case if ϕ is not surjective. (8)

[**25**]

4. Let p be a prime number and S_p the symmetric group of permutations of the set $\{1, \ldots, p\}$.

   (a) Prove that each subgroup of S_p of cardinality p is cyclic and generated by a p-cycle. (5)

   (b) Prove that any two subgroups of S_p with cardinality p are conjugate. (5)

   (c) Compute the number of p-cycles in S_p and the number of subgroups of cardinality p of S_p. (7)

   (d) Let H be a subgroup of S_p with cardinality p and

$$N_{S_p}(H) = \{\sigma \in S_p : \sigma H \sigma^{-1} = H\}$$

   the normaliser of H in S_p.

   By observing the action of S_p by conjugation on the set of its subgroups of cardinality p, or otherwise, show that the cardinality of $N_{S_p}(H)$ equals $p(p-1)$. (8)

[**25**]